EV CHARGING LOAD IMPACT ON A DISTRIBUTION TRANSFORMER AND SUSTAINABLE MICROGRID MITIGATION

Smart Charging, Solar PV, Battery Storage and Vehicle-to-Grid Analysis (VoltGrid Simulator)

Sustainable Energy Engineering (SEE) Project

A PROJECT REPORT

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Under the guidance of

Dr. R. Logudurai

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SUSTAINABLE ENERGY ENGINEERING

in

BACHELOR OF TECHNOLOGY – COMPUTER SCIENCE AND ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this project report titled **"EV CHARGING LOAD IMPACT ON A DISTRIBUTION TRANSFORMER AND SUSTAINABLE MICROGRID MITIGATION"** is the bonafide work of **SHREYAS KUMAR (Register No. RA2311003050085)** and **OMI OJHA (Register No. RA2311003050095)**, students of IV B.Tech Computer Science and Engineering, who carried out the project work for the course **Sustainable Energy Engineering** under my supervision during the academic year 2026-2027. Certified further that, to the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION

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We further declare that this work has not been submitted, in part or in full, to any other university or institution for the award of any degree, diploma or similar title.

Place: Tiruchirappalli
Date: November 2026

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Shreyas Kumar

Omi Ojha

ABSTRACT

The rapid growth of electric vehicles (EVs) adds large, clustered loads to distribution networks that were sized before EVs existed. When charging is uncontrolled, the EV peak coincides with the evening residential peak and can overload neighbourhood transformers, accelerate insulation ageing, depress feeder voltage and raise carbon emissions. This project presents **VoltGrid**, a Python and web-based simulator that quantifies these impacts for a 2.5 MVA, 12.47 kV, 13-bus radial feeder with 70 % EV penetration.

A synthetic dataset of 3,500 EV charging sessions was generated, cleaned to 3,447 validated records and combined with a base load profile and an 800 kWp solar profile. Four strategies were simulated: uncontrolled charging, a static time-of-use tariff, smart valley-filling and a microgrid with solar PV, a 1,200 kWh battery and vehicle-to-grid support. Transformer winding hot-spot temperature and ageing follow the IEEE C57.91 model, feeder voltage and losses follow a LinDistFlow model, and carbon emissions use an hourly marginal intensity profile.

Uncontrolled charging raised the substation peak to 4.46 MW (87.7 % above the 2.375 MW rating), the hot-spot to 189 °C and the ageing rate to 74 times normal. The TOU tariff only shifted the peak to 23:00 (4.31 MW). Smart valley-filling cut the peak to 2.45 MW and the hot-spot to 102 °C, and the microgrid further reduced daily CO_2 by 24.6 % and feeder losses by 30 % relative to uncontrolled charging. The results rely on synthetic data and a single typical day, and the report discusses these limitations. The simulator is supplied with an interactive dashboard (about 2,600 lines of code).

Keywords: Electric Vehicles, Smart Charging, Vehicle-to-Grid, Distribution Transformer Ageing, IEEE C57.91, LinDistFlow, Solar PV, Battery Storage, Sustainable Energy.

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LIST OF ABBREVIATIONS

ANSI: American National Standards Institute; BESS: Battery Energy Storage System; EV: Electric Vehicle; EVSE: EV Supply Equipment; FAA / FEQ: Ageing Acceleration / Equivalent Ageing Factor; IEEE: Institute of Electrical and Electronics Engineers; LOL: Loss of Life; PAR: Peak-to-Average Ratio; PV: Photovoltaic; SoC: State of Charge; TOU: Time of Use; V2G: Vehicle-to-Grid

1. INTRODUCTION

1.1 Background

Transport electrification is one of the main routes to lower carbon emissions, and electric vehicles (EVs) are being adopted rapidly in cities and suburbs. Each vehicle, however, adds a large and highly clustered electrical load to the local distribution network. A single 7.2 kW home charger draws more power than the rest of a typical household, and when many owners plug in after returning from work, the evening residential peak and the EV peak coincide. Neighbourhood distribution transformers and feeders, which were sized for the pre-EV load, are then pushed beyond their ratings.

Overloading shortens the life of the transformer insulation, depresses voltage at the far end of the feeder, and increases line losses. Because the grid often meets the evening peak with gas-fired peaking plants, the added load is also supplied by the most carbon-intensive generation of the day. Whether EV adoption is genuinely sustainable therefore depends not only on how many EVs there are, but on *when* and *how* they charge.

This report describes the **VoltGrid EV Charging Load Impact Simulator**, a software study carried out for the Sustainable Energy Engineering (SEE) course at SRM Institute of Science and Technology, Tiruchirappalli. The study models a 2.5 MVA, 12.47 kV urban-suburban substation and compares four charging strategies, from uncontrolled charging to a solar-plus-storage microgrid with vehicle-to-grid (V2G) support, using a transformer thermal-ageing model and a radial feeder voltage model.

1.2 Motivation

- **Grid reliability:** EV clusters can overload distribution transformers long before generation capacity becomes a problem.
- **Asset life and cost:** hot-spot temperature drives insulation ageing exponentially, so a few hours of overload can cost months of life.
- **Sustainability:** EVs only reduce emissions if they charge when clean energy, such as midday solar, is available.
- **Learning value:** the project combines data engineering, power-system modelling, optimisation logic and an interactive web dashboard.

1.3 Scope of the Project

The work covers synthetic EV charging telemetry and its cleaning, four hourly charging scenarios, an IEEE C57.91 thermal model, a 13-bus LinDistFlow feeder model, an operational carbon estimate and an interactive web dashboard. Network reconfiguration, protection studies, harmonics, battery degradation, tariffs and economic analysis are outside the scope of this version and are listed as future work in Chapter 9.

1.4 Organisation of the Report

Chapter 2 reviews the relevant background. Chapter 3 states the problem and objectives. Chapter 4 describes the test system and data. Chapter 5 presents the methodology and Chapter 6 the implementation. Chapter 7 discusses the results. Chapters 8 to 10 cover limitations, future enhancements and the conclusion.

2. BACKGROUND AND RELATED CONCEPTS

2.1 EV Charging and Distribution Assets

Residential and workplace EVs are usually charged on Level 1 (about 1.4 kW) or Level 2 (7 to 22 kW) AC equipment, while public fast chargers deliver 50 to 150 kW DC. Because most vehicles are parked far longer than the time needed to charge, the charging load is highly *flexible*: the same energy can be delivered at a different time without inconveniencing the driver. Uncontrolled ("dumb") charging ignores this flexibility and starts as soon as the plug is inserted [3].

2.2 Transformer Thermal Ageing (IEEE C57.91)

The life of a liquid-immersed distribution transformer is set by the temperature of its winding insulation. IEEE Std C57.91 [1] relates load, ambient temperature, top-oil temperature and winding hot-spot temperature, and defines an ageing acceleration factor that equals 1.0 at a reference hot-spot temperature of 110 °C. Above this temperature insulation life is consumed at an exponentially increasing rate. Because the oil has a thermal time constant of a few hours, short overloads are partly absorbed, while sustained evening overloads are not.

2.3 Voltage Regulation on Radial Feeders

In a radial feeder, power flowing through the line resistance and reactance causes voltage to fall with distance from the substation. ANSI C84.1 [2] sets the acceptable service voltage at 0.95 to 1.05 per unit (Range A). The LinDistFlow approximation [4] computes bus voltages from the cumulative active and reactive power flowing in each branch and is widely used for planning studies.

2.4 Smart Charging, Solar PV, Storage and V2G

Coordinated ("valley-filling") charging moves flexible demand to hours of low base load [5]. Solar PV supplies energy at midday, and a battery energy storage system (BESS) can absorb surplus solar and release it during the evening peak. Vehicles with bidirectional chargers can additionally inject a small amount of power during the peak (V2G) and recharge overnight [6]. Together these resources form a "sustainable microgrid" that reduces both the peak demand on the transformer and the carbon intensity of the energy used.

3. PROBLEM STATEMENT AND OBJECTIVES

3.1 Problem Statement

Utilities and planners need a simple, transparent way to quantify how much EV charging a neighbourhood transformer and feeder can accept, what damage uncontrolled charging causes, and how much of that damage can be avoided by smart charging and local clean energy. Existing commercial tools are heavy, closed and expensive, while published studies rarely provide code and data that students can run and modify.

3.2 Objectives

1. To build a reproducible data pipeline that generates, cleans and validates EV charging session data.
2. To model four charging strategies for a 2.5 MVA feeder: uncontrolled, static time-of-use (TOU), smart valley-filling and solar + BESS + V2G.
3. To evaluate each strategy using transformer loading, winding hot-spot temperature, ageing, feeder voltage, line losses and CO_2 emissions.
4. To compare the strategies against engineering limits (nameplate rating, 110 °C hot-spot, 0.95 p.u. voltage).

5. To deliver an interactive web dashboard (VoltGrid) so that penetration, solar size and storage size can be varied.

4. SYSTEM DESCRIPTION AND DATA

4.1 Test Feeder and Resources

The test system is a radial 12.47 kV feeder with 13 buses fed from a 2.5 MVA substation transformer. At 0.95 power factor the transformer supplies 2.375 MW continuously, and a short-time emergency limit of 2.65 MW is assumed. Table 1 lists the main parameters.

Table 1: Main system parameters

Item	Value
Substation transformer	2.5 MVA, 12.47 kV; rated 2.375 MW at 0.95 pf; emergency limit 2.65 MW
Feeder	13 buses, 12 branches, 6.5 km in total; r = 0.35 Ω/km, x = 0.38 Ω/km
Base load	Dual-peak profile: 0.71 MW minimum (14:00), 1.52 MW peak (20:00)
Solar PV	800 kWp carport / rooftop array (peak output 800 kW at 12:00 to 13:00)
Battery storage (BESS)	400 kW / 1,200 kWh
EV penetration (default)	70 % of the cleaned session set (2,412 sessions); adjustable from 10 % to 100 %

4.2 Datasets

Four CSV files describe the system (Table 2). The EV session data are synthetic and were generated with a fixed random seed from assumed vehicle mixes (Tesla Model 3 and Y, Ford F-150 Lightning, Hyundai Ioniq 5, Chevrolet Bolt, Nissan Leaf, delivery van) and four user clusters: residential commuters (65 %), workplace commuters (20 %), commercial fleet (10 %) and public fast charging (5 %). About 3.8 % of the raw records were deliberately corrupted to mimic real telemetry faults.

Table 2: Datasets used in the study

File	Records	Purpose
raw_ev_charging_data.csv	3,500	Unfiltered sessions with injected noise, missing values and spikes
cleaned_ev_charging_data.csv	3,447	Validated and feature-engineered sessions used for simulation
distribution_grid_hourly_load.csv	24	Hourly base load (MW) and ambient temperature
solar_generation_profile.csv	24	Hourly PV capacity factor and output of the 800 kWp array

4.3 Data Preprocessing

The pipeline (`data_pipeline.py`) removed sessions whose departure was not after arrival (dwell shorter than 9 minutes), imputed missing initial and target state of charge (SoC) with medians for each user type, rejected power readings above 350 kW, replaced zero energy values using battery physics, and enforced a target SoC above the initial SoC. Of the 3,500 raw records, 3,447 (98.5 %) were retained. Energy per session was computed as $E = (SoC_{target} - SoC_{initial}) / (100 \times \eta) \times C_{battery}$ with a charging efficiency $\eta = 0.90$, and a *charging flexibility index* $\Phi = t_{dwell} / t_{min}$ was added. The cleaned data have a mean energy request of 41.1 kWh, a mean dwell of 10.1 h, and 89 % of sessions have Φ above 1, meaning that most charging can be shifted in time.

5. METHODOLOGY

5.1 Overall Workflow

Figure 1 shows the workflow. Cleaned sessions are converted into hourly EV load under each strategy and added to the base load. The resulting net load of the substation is passed to the transformer and feeder models, and the metrics are exported for plotting and for the dashboard.

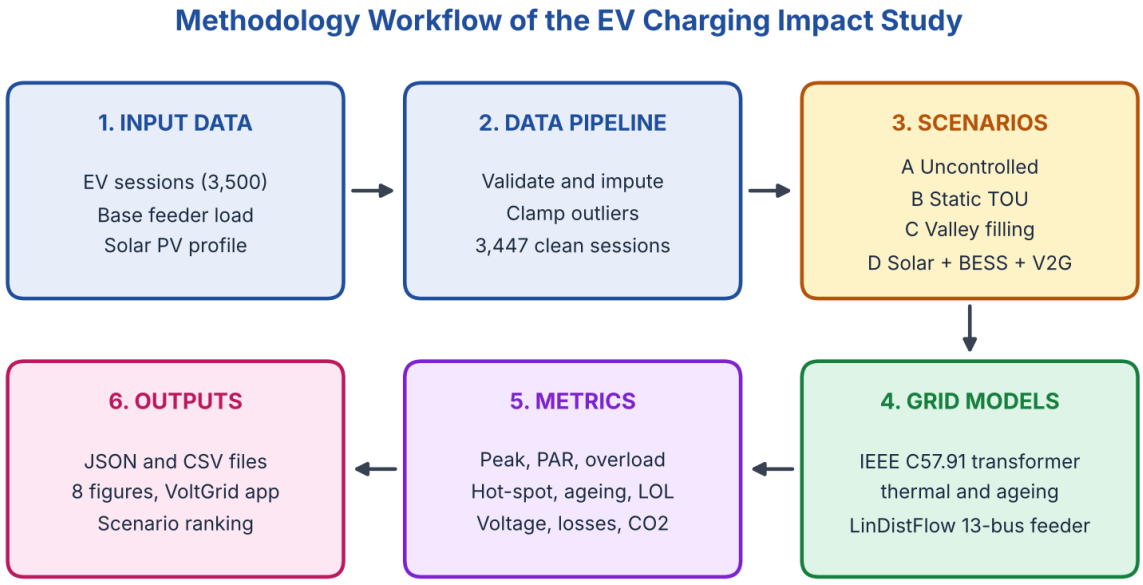


Figure 1: Methodology workflow of the study

5.2 Charging Scenarios

Table 3: Charging scenarios compared

Scenario	Description
Baseline	Base feeder load with no EVs; reference case.
A. Uncontrolled	Each EV charges at full charger power from the hour of arrival until its energy request is met.
B. Static TOU tariff	Residential commuters delay charging to 23:00 (start of off-peak price); all other users charge on arrival.
C. Smart valley-filling	22 % of the EV energy is non-deferrable; the remaining 78 % is allocated to hours in inverse proportion to the base load, with 17:00 to 22:00 suppressed.
D. Microgrid (solar + BESS + V2G)	Scenario C plus 800 kWp PV; BESS charges from solar (10:00 to 14:00, up to 350 kW) and discharges 320 kW at 18:00 to 21:00; V2G injects up to 0.22 MW × eligible fraction at 18:00 to 20:00 and recharges overnight.

5.3 Transformer Thermal and Ageing Model

The hourly load is expressed as a per-unit ratio $K = P / P_{rated}$. The ultimate top-oil rise and the hot-spot rise follow IEEE C57.91 [1], and the top-oil temperature is advanced in one-hour Euler steps:

$$\Delta\theta_{TO,U} = \Delta\theta_{TO,R}\,[(K^2R + 1)/(R + 1)]^n$$

$$\theta_{TO}(t) = \theta_{TO}(t-1) + (\Delta t/\tau_{TO})\,[\theta_{TO,U} - \theta_{TO}(t-1)]$$

$$\theta_H = \theta_{TO} + \Delta\theta_{H,R}\,K^{2m}$$

$$F_{AA} = \exp[15000/383 - 15000/(\theta_H + 273)];\quad F_{EQ} = \mathrm{mean}(F_{AA});\quad LOL\% = F_{EQ} \times (24/180000) \times 100$$

The parameters are $\Delta\theta_{TO,R}$ = 55 °C, $\Delta\theta_{H,R}$ = 25 °C, τ_{TO} = 3 h, R = 5 and n = m = 0.8. Normal insulation life is taken as 180,000 hours, so a day at F_{EQ} = 1 consumes about 0.0133 % of life. Ambient temperature varies from 16 °C at night to 35 °C in the afternoon.

5.4 Feeder Voltage and Loss Model

Total substation power is shared among buses 2 to 13 using fixed load shares (for example 12 % at the commercial centre, bus 6). A backward sweep accumulates branch flows P_i and Q_i (0.95 pf lagging), and a forward sweep applies the LinDistFlow voltage drop:

$$V_{i+1} = V_i - (r_i P_i + x_i Q_i)/V_i; \quad P_{loss,i} = [(P_i^2 + Q_i^2)/V_i^2]\, r_i$$

5.5 Carbon Accounting

Daily operational emissions are computed as the sum over 24 hours of net substation load multiplied by an hourly marginal carbon intensity, which ranges from 90 kg/MWh at midday (solar) to 520 kg/MWh at 18:00 (gas peakers). The intensity profile is an assumption made in the simulation code and not a measured grid value.

6. IMPLEMENTATION

6.1 Project Structure

The project is written in Python 3 (NumPy, pandas, Matplotlib) for the simulation and in HTML, CSS and JavaScript (Chart.js) for the dashboard.

```
see-project/
|-- datasets/    data_pipeline.py, raw and cleaned EV data, grid load, solar profile
|-- simulation/  smart_charge_optimizer.py, transformer_thermal_model.py,
|                grid_feeder_model.py, run_simulation.py, generate_plots.py,
|                simulation_summary.json, scenario_hourly_timeseries.csv
|-- figures/     8 PNG figures (load, hot-spot, voltage, demand, dispatch, ...)
`-- app/         index.html, styles.css, app.js (VoltGrid dashboard)
```

Table 4: Source file sizes

File	Role	Lines
data_pipeline.py	Data generation, cleaning, grid and solar profiles	236
smart_charge_optimizer.py	Scenario A to D dispatch logic	174
transformer_thermal_model.py	IEEE C57.91 thermal and ageing model	94
grid_feeder_model.py	13-bus LinDistFlow voltage and loss model	122
run_simulation.py, generate_plots.py	Orchestration, metrics export and figures	105 + 337
index.html, styles.css, app.js	VoltGrid web dashboard	297 + 675 + 560
Total		**2,600**

6.2 VoltGrid Web Dashboard

The dashboard re-implements the scenario, thermal, voltage and carbon calculations in JavaScript so that results update instantly in the browser. The user selects a charging strategy and moves sliders for EV penetration (10 to 100 %), solar capacity (0 to 1,500 kWp) and storage (0 to 2,400 kWh). Four KPI cards show peak load, hot-spot temperature with the ageing multiple, minimum feeder voltage and daily CO_2. Charts show the 24-hour net load against the transformer limits and the bus voltage profile, a table compares reliability metrics, a "Run Monte Carlo" button steps the penetration rate through random values between 40 % and 95 %, and results can be exported as CSV. A filterable table lists the simulated EV fleet and its flexibility.

7. RESULTS AND DISCUSSION

All results are for the default case of 70 % EV penetration, 800 kWp PV and a 1,200 kWh BESS. Table 5 summarises the metrics of all five cases.

Table 5: Summary of simulation results (24-hour day)

Metric	Baseline	A. Uncontrolled	B. Static TOU	C. Valley filling	D. Microgrid
Peak load (MW)	1.52	4.46	4.31	2.45	2.45
Peak-to-average ratio	1.46	2.13	2.06	1.17	1.26
Above 2.375 MW rating (%)	0.0	87.7	81.6	3.2	3.2
Max hot-spot temp (°C)	63.6	189.4	176.2	102.2	97.0
Ageing factor F_{EQ}	0.002	74.190	24.110	0.193	0.095
Daily loss of life (%)	0.00002	0.98921	0.32147	0.00258	0.00127
Min feeder voltage (p.u.)	0.9836	0.9511	0.9527	0.9734	0.9734
Feeder losses (MWh/day)	0.175	0.847	0.790	0.679	0.596
CO_2 (kg/day)	6,761	14,868	11,764	12,176	11,206

7.1 Substation Load

Without EVs the base load peaks at 1.52 MW, well below the rating. Uncontrolled charging adds an EV peak that coincides with the evening residential peak, giving 4.46 MW at 20:00, which is 87.7 % above nameplate and above the 2.65 MW emergency limit for five hours. The static TOU tariff does not remove the problem: because residential users all start at 23:00, a rebound peak of 4.31 MW appears at 23:00 and the transformer exceeds its rating for eight hours. Smart valley-filling cuts the peak to 2.45 MW (a 45 % reduction from Scenario A) and the peak-to-average ratio from 2.13 to 1.17.

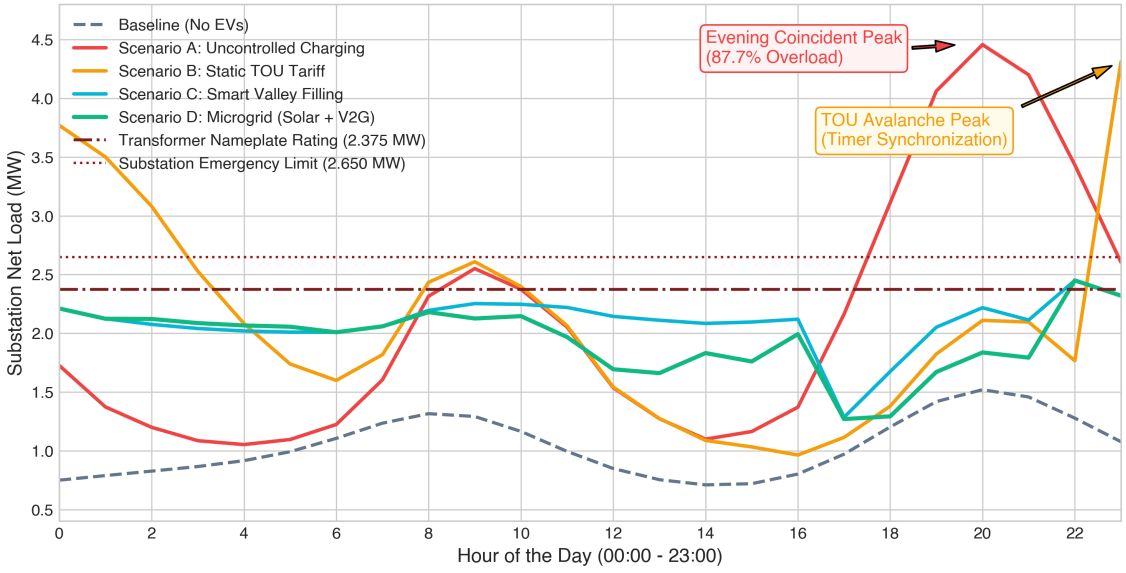


Figure 2: 24-hour substation net load for the four charging scenarios

7.2 Transformer Temperature and Ageing

Uncontrolled charging drives the winding hot-spot to 189 °C, far above the 110 °C reference, and the ageing factor reaches 74.2, so the transformer loses about 0.99 % of its life in one day, roughly 74 times the normal rate (0.0133 %). The TOU case is still severe (176 °C, F_{EQ} = 24.1). Smart valley-filling holds the hot-spot at 102 °C and the microgrid at 97 °C, both below 110 °C, with F_{EQ} of 0.19 and 0.095, i.e. the transformer ages well below the normal rate (F_{EQ} < 1), although still above the no-EV baseline value of 0.002.

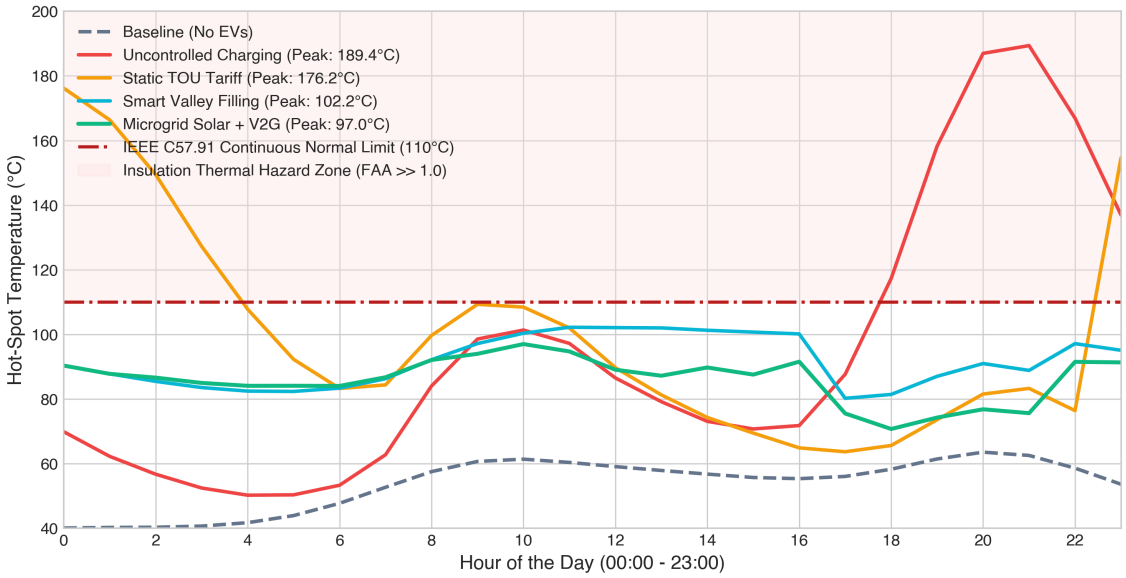


Figure 3: Winding hot-spot temperature against the 110 °C limit

7.3 Feeder Voltage and Losses

At the evening peak the voltage at the feeder tail (bus 13) falls to 0.9511 p.u. under uncontrolled charging and 0.9527 p.u. under TOU, just above the 0.95 p.u. limit of ANSI C84.1, so no hour violates the limit in the default case. Smart charging improves the minimum voltage to 0.9734 p.u. Feeder losses fall from 0.847 MWh/day (A) to 0.679 (C) and 0.596 MWh/day (D), a 30 % cut for the microgrid. The margin under uncontrolled charging is thin: a higher penetration rate would be expected to cross the limit, which the user can examine by raising the penetration slider in the dashboard.

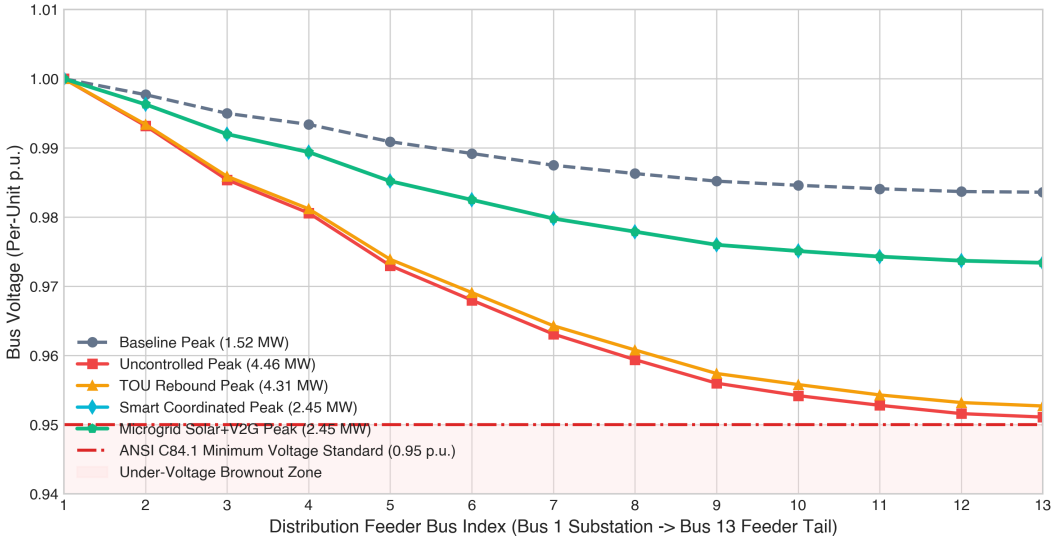


Figure 4: Bus voltage profile along the feeder at peak loading

7.4 Solar, Storage and V2G Dispatch

Figure 5 shows the microgrid dispatch. Solar output peaks at 800 kW at midday and charges the BESS between 10:00 and 14:00; the BESS then discharges 320 kW from 18:00 to 21:00, and V2G adds a small injection at 18:00 to 20:00. The evening load therefore falls markedly. The remaining maximum of 2.45 MW occurs at 22:00, just after the discharge window ends, which shows that the dispatch window, not the storage size alone, limits the peak.

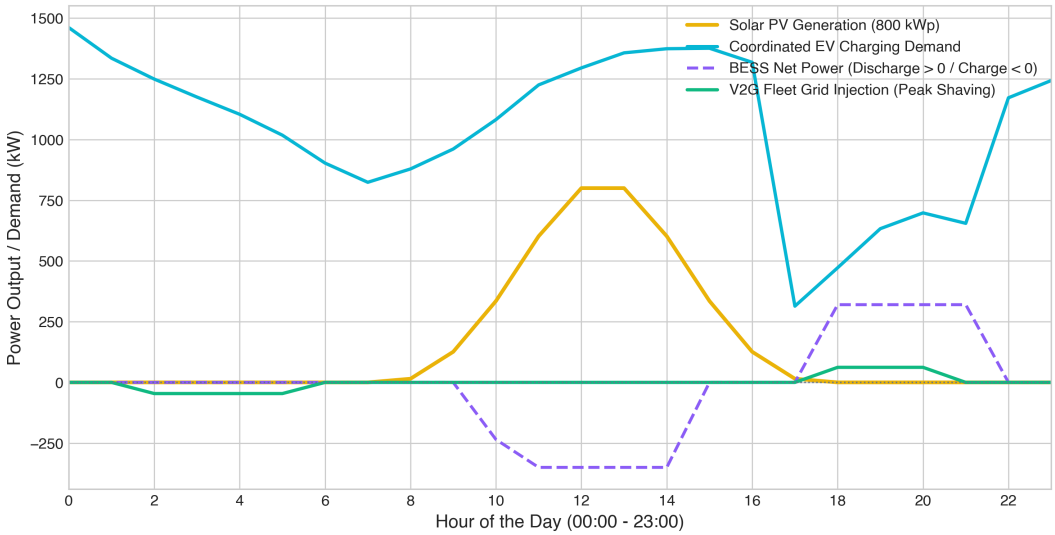


Figure 5: Microgrid dispatch of solar PV, BESS and V2G

7.5 Carbon Emissions

Uncontrolled charging more than doubles the daily emissions compared with the no-EV baseline (14,868 against 6,761 kg/day) because the extra energy is supplied at the high-carbon evening hours. The microgrid reduces emissions to 11,206 kg/day, avoiding 3,662 kg of CO_2 per day (24.6 %). An important observation is that smart valley-filling (12,176 kg) emits slightly more than the static TOU case (11,764 kg): flattening the load protects the transformer but does not by itself minimise carbon, because the shifted energy lands in night hours whose assumed marginal intensity (150 to 240 kg/MWh) is higher than at midday. Carbon-aware scheduling is therefore a separate objective from peak shaving.

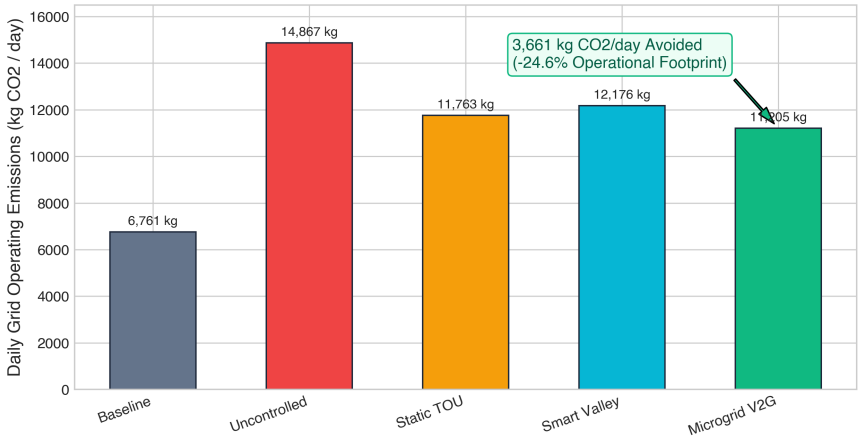


Figure 6: Daily operational CO_2 emissions by scenario

7.6 Achievement of Objectives

Table 6: Objectives versus outcome

No.	Objective	Outcome
1	Reproducible data pipeline	Achieved: 3,500 raw to 3,447 clean sessions with documented rules
2	Four charging strategies modelled	Achieved: Scenarios A to D plus baseline
3	Thermal, voltage, loss and CO_2 metrics	Achieved: 14 metrics per scenario exported to JSON and CSV
4	Comparison against limits	Partly: C and D keep the hot-spot below 110 °C but peak at 2.45 MW, 3.2 % above nameplate
5	Interactive dashboard	Achieved: VoltGrid with three sliders, KPI cards, charts and CSV export

8. LIMITATIONS

- **Synthetic data:** EV sessions, base load, solar and ambient temperature are generated, not measured, so the numbers illustrate relative behaviour and are not a forecast for a real feeder.
- **Single typical day:** only one 24-hour day at hourly resolution is simulated; seasons, weekends and sub-hourly peaks are not covered.
- **Simplified network:** a balanced, single-phase-equivalent radial feeder with fixed load shares and LinDistFlow approximation; no unbalance, taps or capacitor control.
- **Heuristic scheduling:** valley-filling uses fixed weights and a 22 % non-deferrable share, not a formal optimisation with driver constraints.
- **Assumed carbon intensity and fixed V2G, BESS rules:** no battery degradation, round-trip losses or economics are modelled.
- **Residual overload:** Scenarios C and D still reach 2.45 MW (3.2 % above nameplate) for one hour, within the 2.65 MW emergency limit.

9. FUTURE ENHANCEMENTS

- Replace synthetic data with real charger logs and smart-meter data from a utility or campus.
- Use linear or mixed-integer optimisation for charging with driver departure-time constraints and a carbon-aware objective.
- Add multi-day and seasonal simulation, three-phase unbalanced power flow and voltage-regulation devices.
- Include BESS degradation, V2G battery wear, tariffs and a payback analysis.
- Extend the dashboard with uploads of user data and a rolling forecast of load and solar.

10. CONCLUSION

The VoltGrid study shows, within the stated assumptions, that EV charging on an existing 2.5 MVA feeder is manageable only if it is coordinated. At 70 % penetration, uncontrolled charging raises the peak to 4.46 MW, heats the winding to 189 °C and ages the transformer 74 times faster than normal, and a simple TOU tariff merely moves the peak to 23:00. Smart valley-filling reduces the peak by 45 % and removes the thermal hazard, and adding solar, storage and V2G lowers the daily CO_2 by 24.6 % and feeder losses by 30 % compared with uncontrolled charging. The results also show that peak shaving and decarbonisation are different goals and need to be designed together. The code, data and dashboard form a reproducible base for future work on real data and optimised control.

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